

How are we implementing Gradient Descent?

↑ Optimize

GD → Batch GD → memory issue → epoch → update

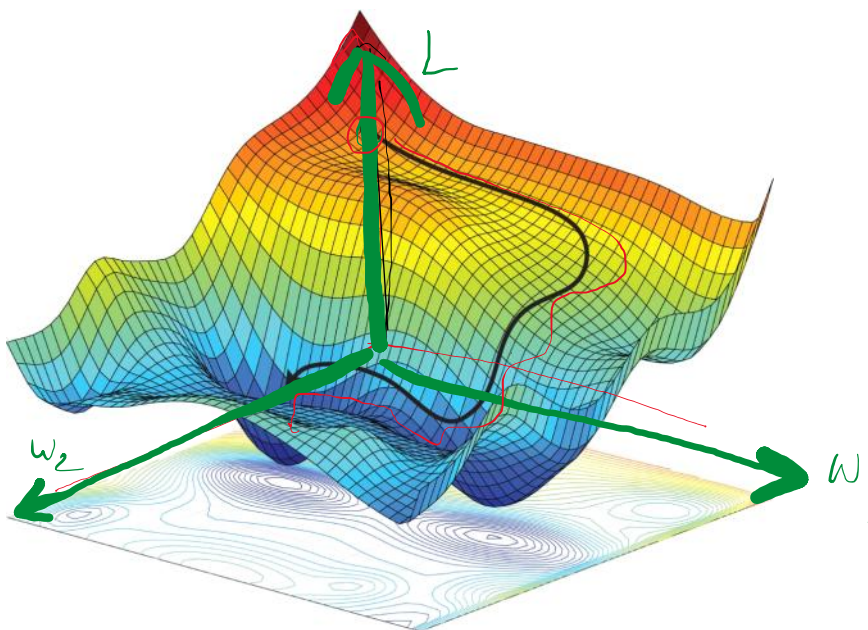
→ Stochastic GD → slow → update

→ Mini Batch GD → Unstable training

→ Batch

$$W = W - \eta \frac{dL}{dW}$$

Fast / slow we want move



$$L = f(w_1, w_2)$$

η → How fast / slow we want to move

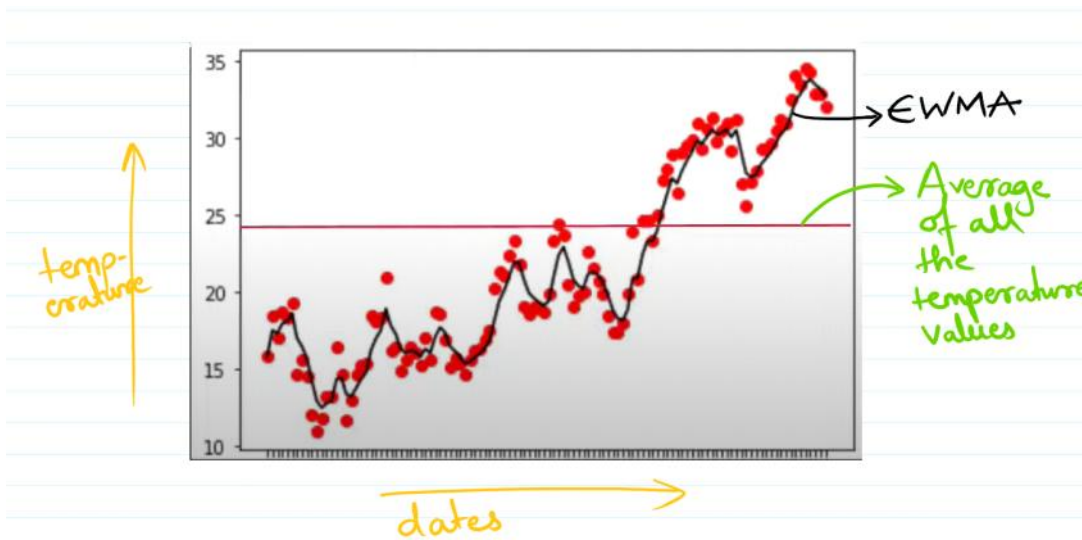
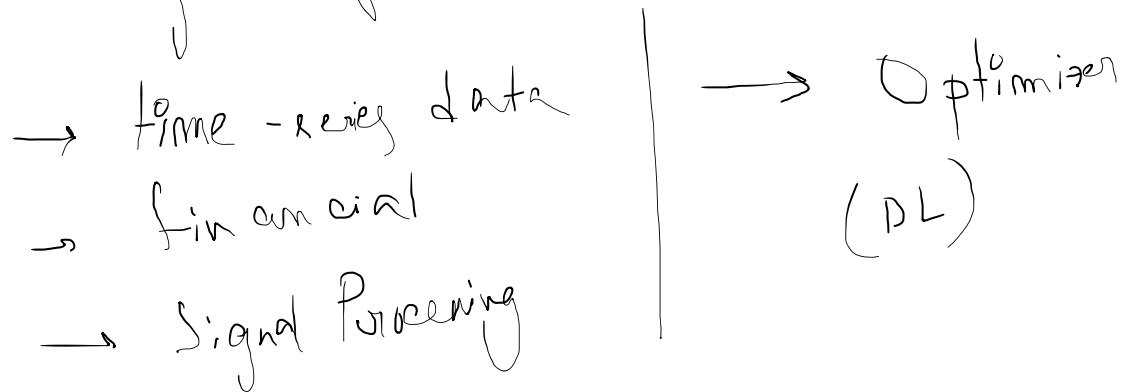


X Learning Rate Scheduling:-

Pre defined $\rightarrow$ [Small $\eta \rightarrow$ increase η]
 $\nwarrow$
Not data dependent

EWMA ✓
Momentum
NAG
RMSProp
Adam ✓
 $\nwarrow$
used

Exponentially Weighted Moving Average



$$V_t = \beta V_{t-1} + (1-\beta) \theta_t$$

data Attribute

EWMA at time t

$$0 < \beta < 1$$

$$0.9$$

$$\beta = 0.9$$

EWMA at $t=0$

$$V_0 = 0 \quad | \quad V_0 = \theta_0$$

$$V_0 = 25$$

$$\frac{\text{index}}{D_1}$$

$$D_2$$

$$\frac{\text{temp}(\theta)}{25}$$

$$13$$

$$V_1 = 0.9 * 25 + \frac{D_3}{(1-0.9) \times 25} \quad 1.7$$

$$V_1 = 1.3$$

$$V_2 = 0.9 * 1.3 + (1-0.9) * 13 \Rightarrow V_2 = 0.9 * V_1 + 1.3$$

$$V_3 = 0.9 * V_2 + (1-0.9) * 17$$

$$V_3 = 0.9 V_2 + 1.7$$

$$D_4 > D_3 > D_2 > D_1$$

$$\beta = V_t = \leftarrow \text{last}$$

$$\frac{1}{1-\beta}$$

intuition

$$\beta = 0.9$$

$$\frac{1}{1-0.9} = \frac{1}{0.1} = 10 \leftarrow \text{last 10 days average}$$

$$\beta = 0.98$$

$$\frac{1}{1-0.98} = \frac{1}{0.02} = 50 \text{ days}$$

$$\beta = 0.5$$

$$\frac{1}{1-0.5} = \frac{1}{0.5} = 2 \text{ days}$$

$$D_4 > D_3 > D_2 > D_1 ?$$

$$V_t = \beta V_{t-1} + (1-\beta)\theta_t$$

$$V_0 = 0$$

$$V_1 = (1-\beta)\theta_1$$

$$V_2 = \beta[(1-\beta)\theta_1] + (1-\beta)\theta_2$$

$$V_3 = \beta V_2 + (1-\beta)\theta_3$$

$$= \beta[\beta(1-\beta)\theta_1 + (1-\beta)\theta_2] + (1-\beta)\theta_3$$

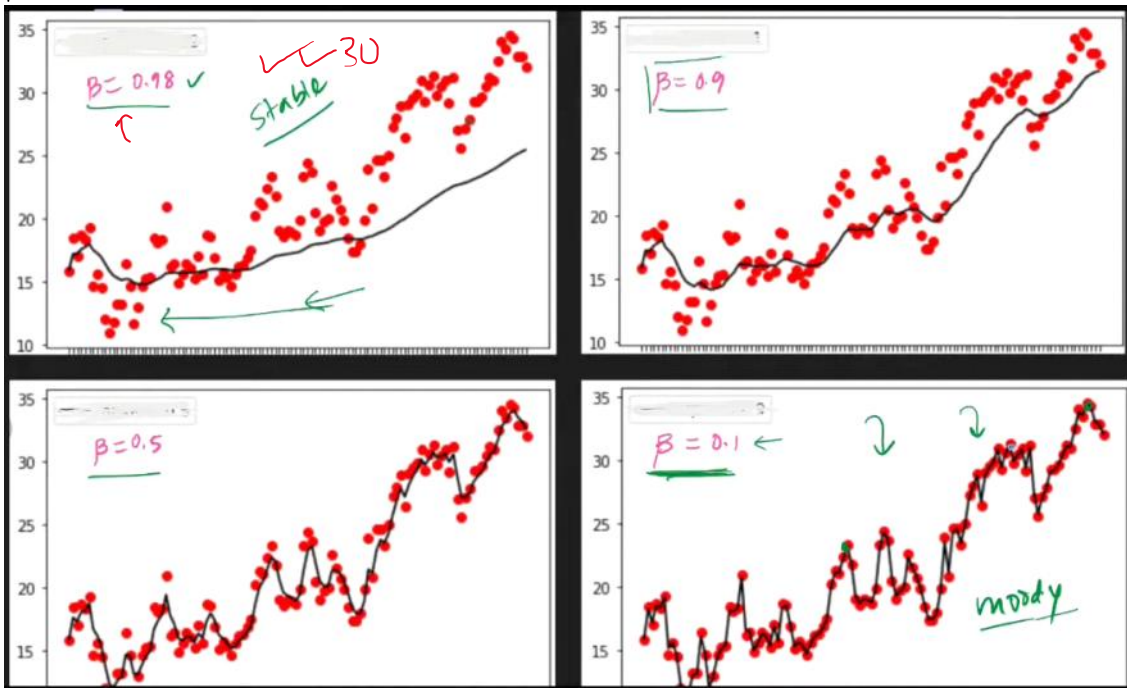
$$V_3 = \beta^2(1-\beta)\theta_1 + \beta(1-\beta)\theta_2 + (1-\beta)\theta_3$$

$$V_4 = \beta^3(1-\beta)\theta_1 + \beta^2(1-\beta)\theta_2 + \beta(1-\beta)\theta_3 + (1-\beta)\theta_4$$

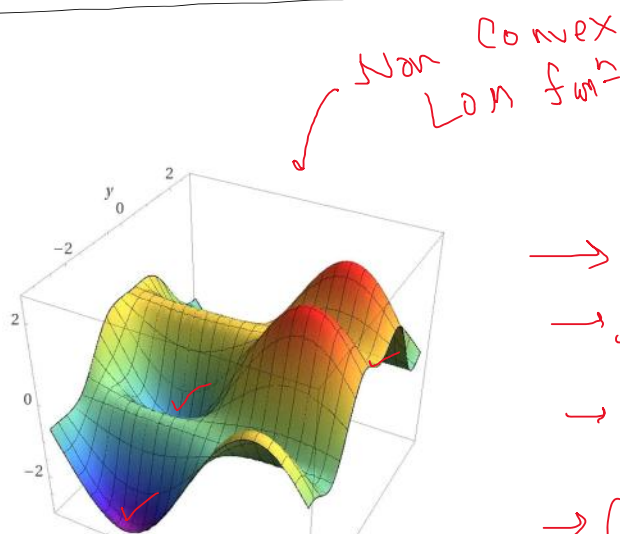
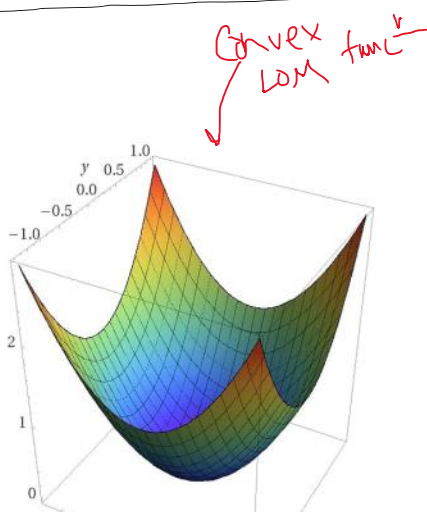
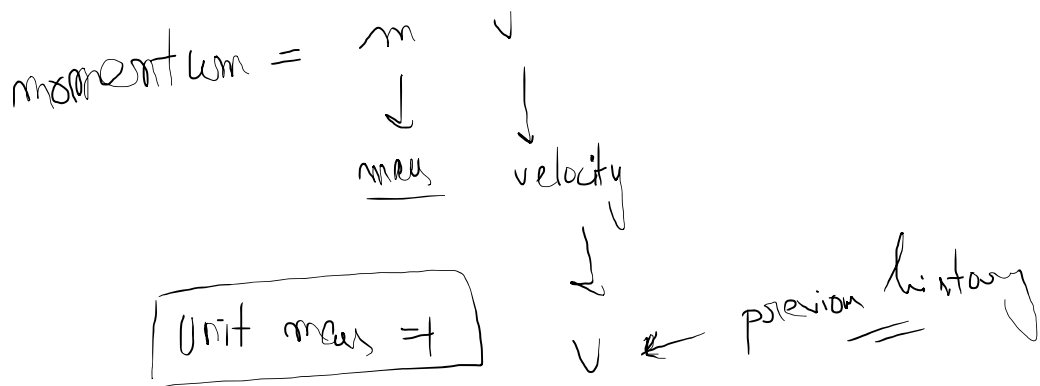
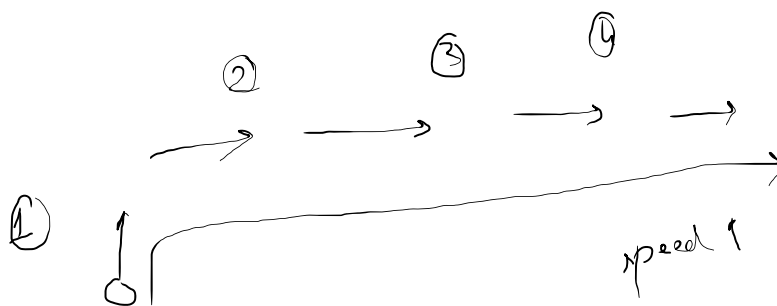
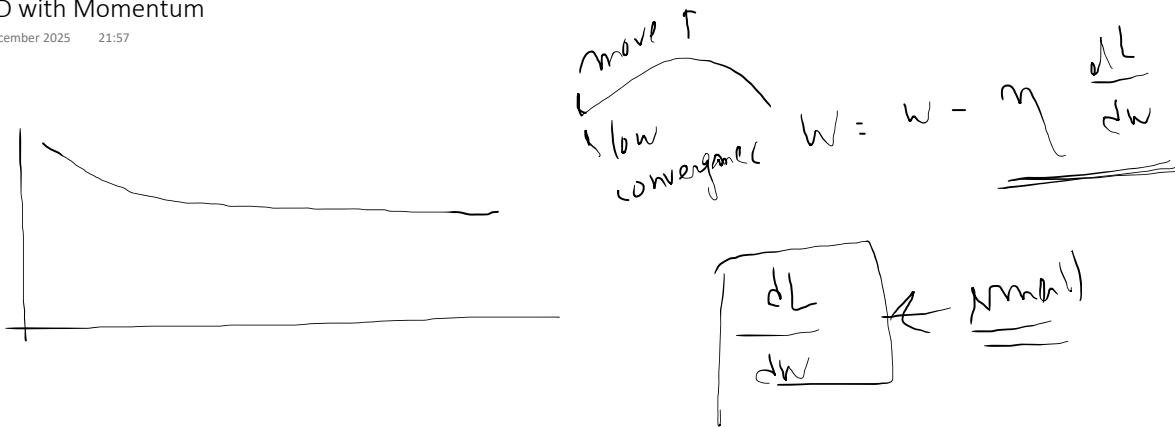
$$V_n = \frac{(1-\beta)}{\beta} \left[\frac{\beta^3\theta_1}{\beta} + \frac{\beta^2\theta_2}{\beta} + \frac{\beta\theta_3}{\beta} + \frac{\theta_4}{\beta} \right]$$

$$0 < \beta < 1$$

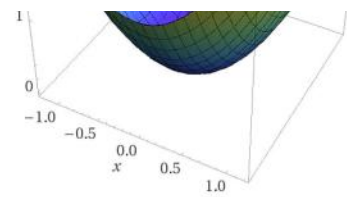
$$V_t = (1-\beta) \left[\beta^{t-1}\theta_1 + \beta^{t-2}\theta_2 + \dots + \beta\theta_{t-1} + \theta_t \right]$$



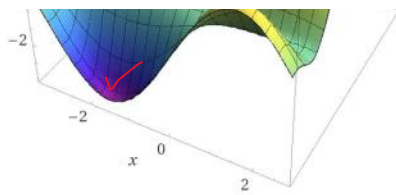
$$\frac{1}{-0.1} = \frac{1}{0.9} = \frac{10}{9} = \boxed{1.1}$$



- Local Minima
- Saddle point
- High Curvature
- Consistent

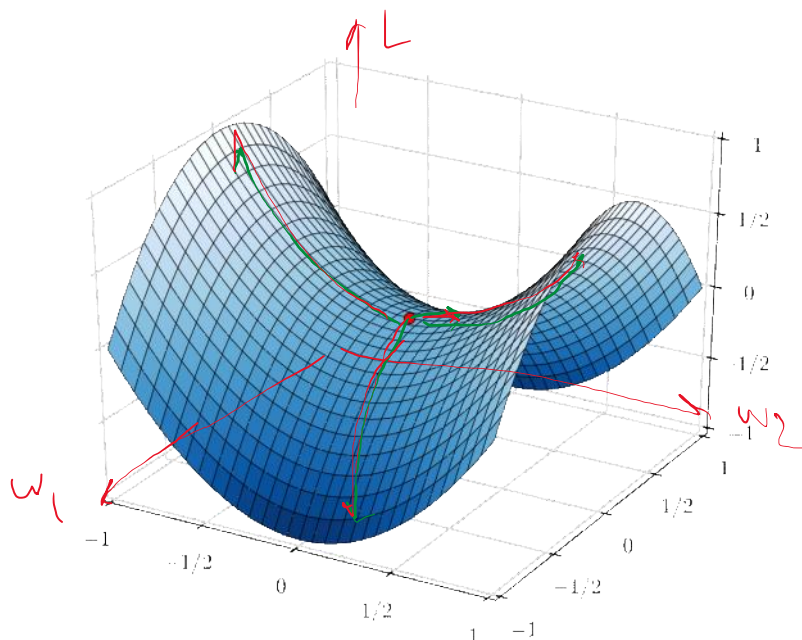
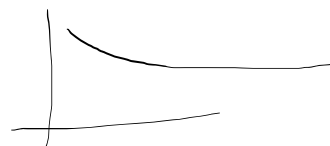


Computed by WolframAlpha

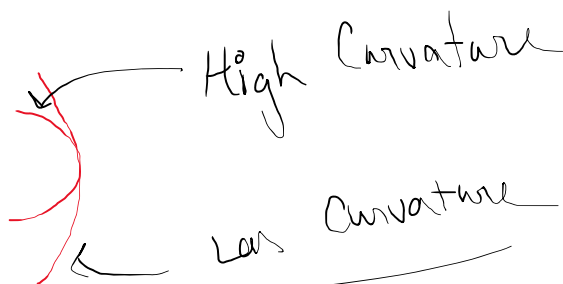


Computed by WolframAlpha

$\rightarrow$ Constant gradient



Saddle point



SGD with momentum $\gg$ SGD (99.9%)

$$w_{t+1} = w_t - \eta \Delta w_t \leftarrow \text{SGD}$$

EWMA at time t

EWIHT ~

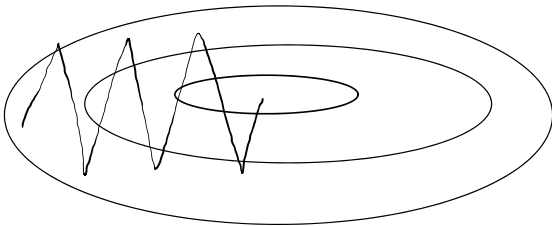
$$\Rightarrow \boxed{w_{t+1} = w_t - v_t}$$

$$v_t = \beta * v_{t-1} + \eta \Delta w_t$$

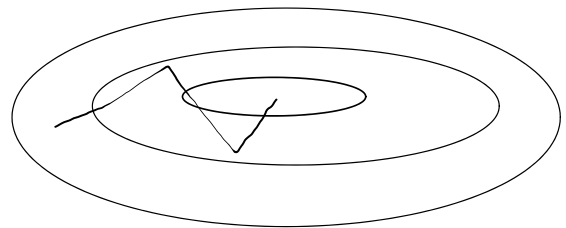
$$\boxed{w_{t+1} = w_t - (\beta v_{t-1} + \eta \Delta w_t)}$$

← SGD
with momentum

$$\beta=0 \Rightarrow w_{t+1} = w_t - \eta \Delta w_t \quad (\text{SGD})$$



(without momentum)



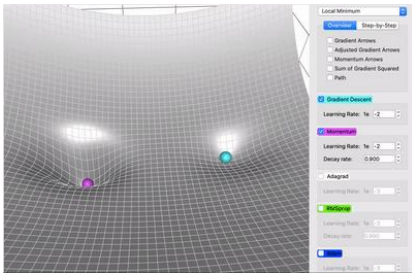
Higher Horizontal movement
(with momentum)

$\beta \leftarrow$ decaying factor

$$\beta_1 = 1 \quad \boxed{w_{t+1} = w_t - \nabla_{t-1} - \eta \Delta w_t}$$

↑ dynamic equilibrium / no fading

$$\boxed{0 < \beta < 1} \quad \beta = 0.9$$



<https://contributor.insightmediagroup.io/wp-content/uploads/2025/03/image-21.gif>

advantage
momentum
→ fast convergence

disadvantage
momentum
→ unnecessary oscillations

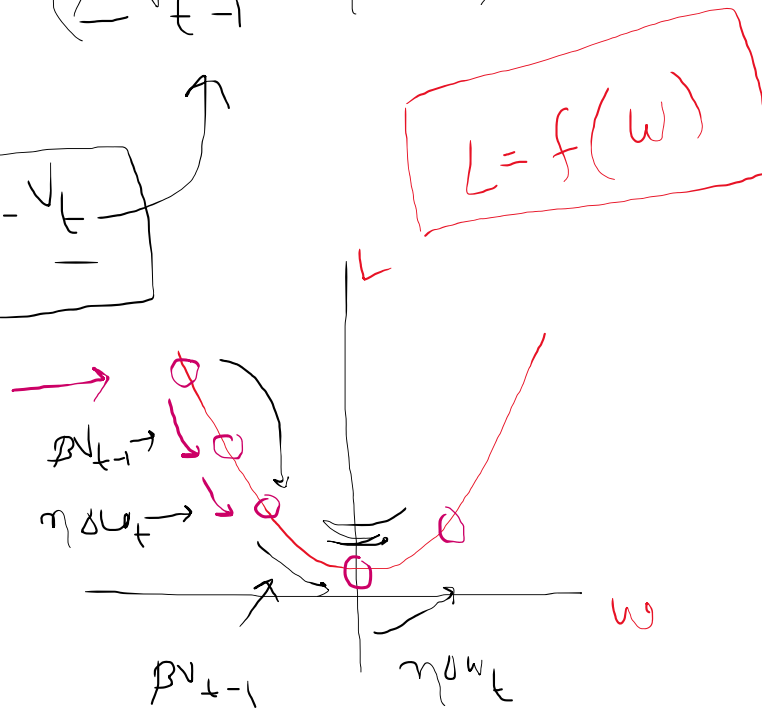
↑ Nesterov Accelerated Gradient

⊗ Problems in Momentum: -

$$w_{t+1} = w_t - (\beta \nabla_{t-1} + \eta \Delta w_t)$$

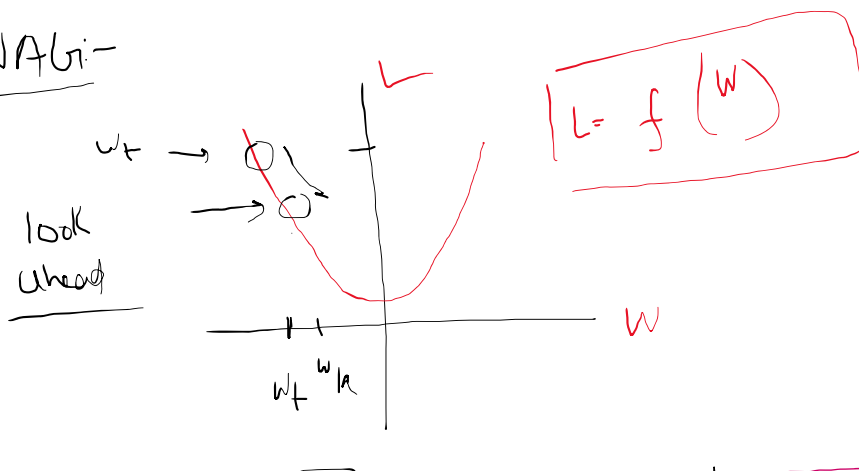
$$w_{t+1} = w_t - \nabla_t$$

- ① $\beta \nabla_{t-1}$
- ② $\eta \Delta w_t$



- i) Calculating $\beta \nabla_{t-1}$ at time t
- ii) Calculating $\eta \Delta w_t$ at time t

⊗ NAG:-

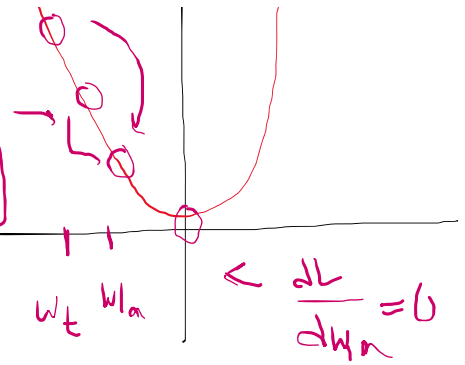


w_t is

$$w_{la} = (w_t - \beta V_{t-1})$$

← lookahead

$$\frac{dL}{dw_{la}}$$



$$V_t = \beta V_{t-1} + \eta \Delta w_{la}$$

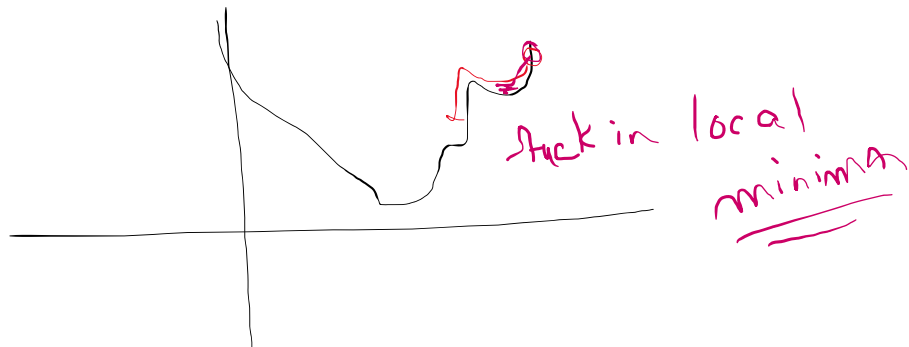
✓

$$w_{t+1} = w_t - V_t$$

✓

1) disadvantage

→ Oscillation is being damped in



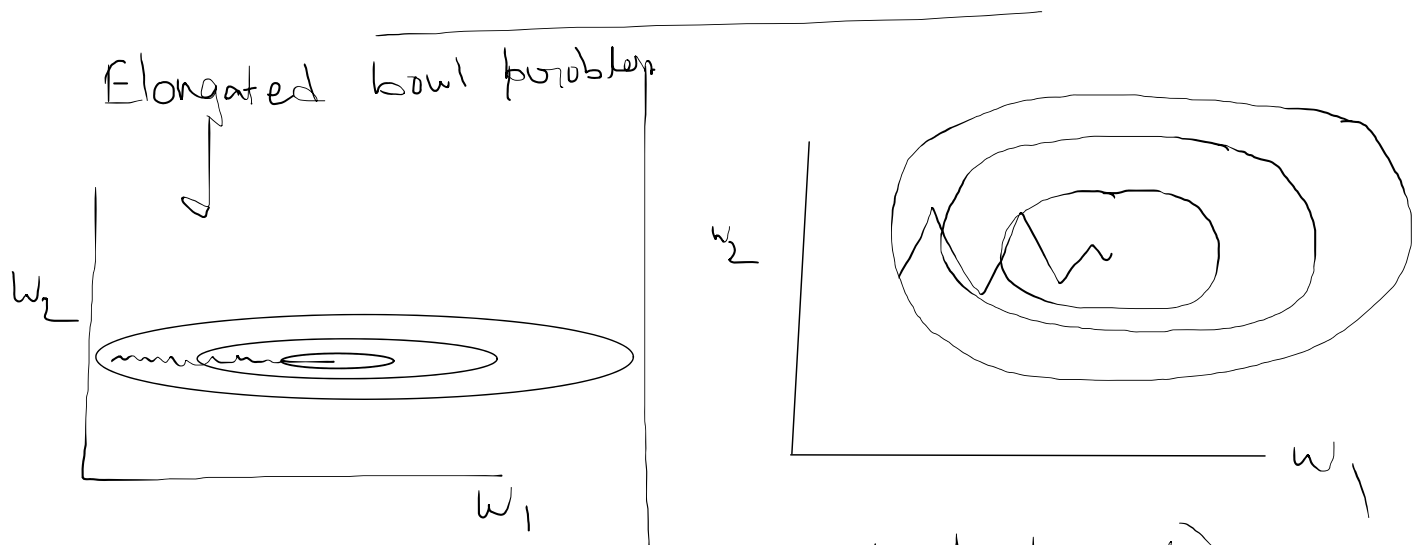
when to we $\rightarrow$ for diffⁿ scaled i/p

<u>Age</u>	<u>Salary</u>	<u>loan (Y/N)</u>
$(0-100)$	$(10k-1Cr)$	

$\swarrow$ Scaling (fixable)

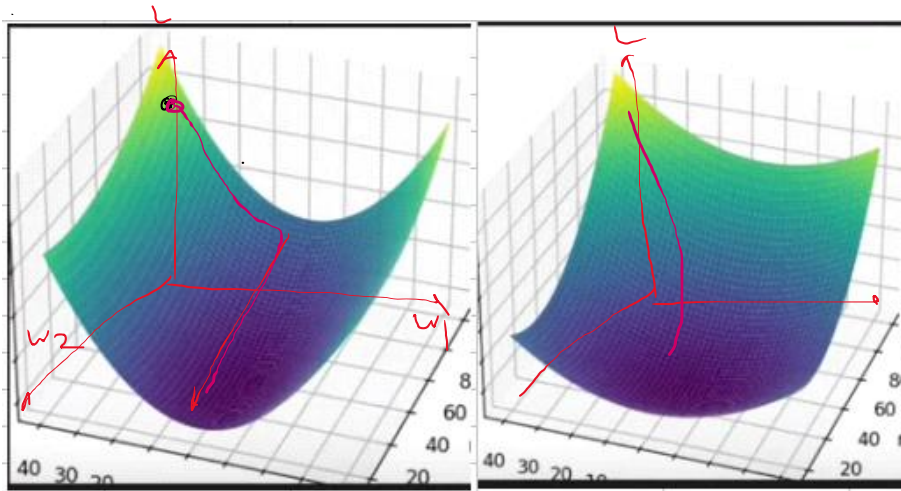
$\rightarrow$ Sparse feature $\rightarrow$ most of the values are zero

w_1	<u>iq</u>	<u>farm_iiit</u> w_2	<u>placement</u>
	051	0	Y
	109	0	N
	75	0	N
	112	1	Y
	100	0	Y
	67	0	N



(with space feature) w_1

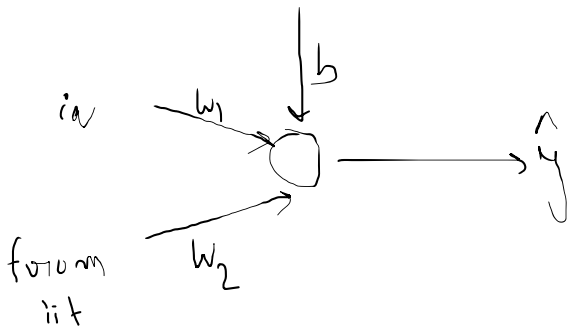
(without space) w_1



✓ Support Regⁿ

$$\hat{y} = w_1 x_1 + w_2 x_2 + b$$

$$= w_1 i_q + w_2 f_{it} + b$$



$$L = (y - \hat{y})^2$$

$$w_1 = w_1 - \eta \frac{dL}{dw_1}$$

$$\frac{dL}{dw_1} = -2(y - \hat{y}) i_q$$

$$w_2 = w_2 - \eta \frac{dL}{dw_2}$$

$$\frac{dL}{dw_2} = -2(y - \hat{y}) \frac{f_{it}}{\uparrow}$$

mostly 0

$$\frac{dL}{dw_2} = 0 \text{ (mostly)}$$

$$w_2 = w_2 - \eta \frac{dL}{dw_2}$$

$$= w_2 - \eta \times 0$$

$$\frac{dL}{dw_2} = 0 \text{ (mostly)}$$

$$= w_2 - \eta \times 0$$

$$= w_2 - 0 = \underline{w_2}$$

$$\left[\begin{array}{l} w_1 \leftarrow \text{big update} \\ w_2 \leftarrow \text{very small update} \end{array} \right]$$

slow training

Soln:

softmax is sparse feature (mostly values = 0)

when I get non zero values, all? X

$$\left[\frac{dL}{dw_2} \right] \rightarrow \text{non zero}$$

given $\text{update}(w_1) \approx \text{update}(w_2)$

slow

fast

$\eta \leftarrow$ learning rate big

very small

Adagrad

✓

$$w_1 = \underline{\underline{w_1 - \eta \delta w_t}}$$

and when

$$w_{t+1} = w_t - \frac{\eta \Delta w_t}{\sqrt{V_t + \epsilon}} \quad \leftarrow \text{epsilon term (used when } V_t = 0)$$

$$V_t = V_{t-1} + (\Delta w_t)^2 \quad \leftarrow \text{storing past gradients squared sum}$$

if $\frac{1}{\sqrt{V_{t+k}}} \rightarrow \text{big} \Rightarrow \frac{\eta}{\sqrt{V_{t+k}}} \rightarrow \text{very small}$

if $\frac{1}{\sqrt{V_{t+k}}} \rightarrow \text{small} \Rightarrow \frac{\eta}{\sqrt{V_{t+k}}} \rightarrow \text{no change}$

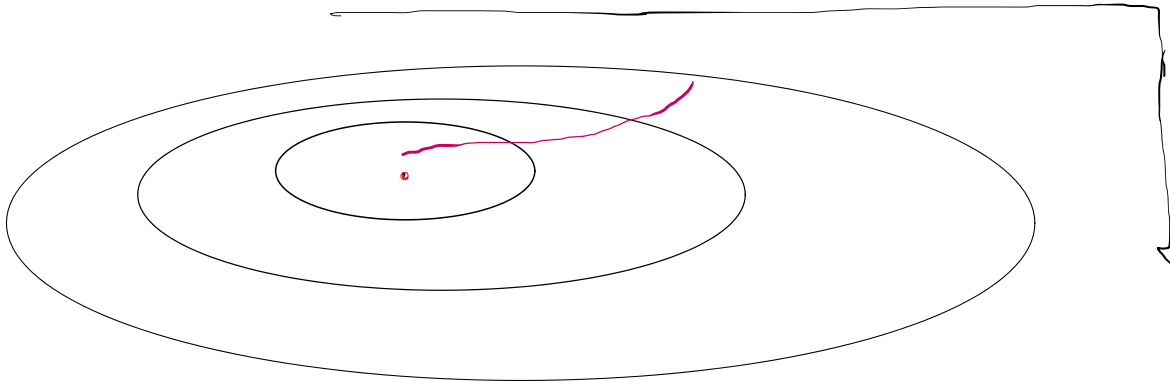
$$\frac{5}{8} = 0.63$$

$$\frac{5}{0.01} \rightarrow 500$$

Disadv:-

It never does converge.
for any no. of epochs

Never used



gradually $V_t \rightarrow \uparrow \rightarrow \eta \downarrow$
 V_t ^{squared} containing the summation of all prev Δw_t

$$\frac{\eta}{\sqrt{V_t + \epsilon}} \Delta w_t \approx 0$$

$$\boxed{w_{t+1} = w_t} \leftarrow \text{no learning}$$

RMSProp $\rightarrow$ Adagrad + Exponentially
decaying
gradients

$$W_{t+1} = W_t - \frac{\eta}{\sqrt{V_{t+1}}} \Delta W_t$$

$$V_t = \beta V_{t-1} + (1-\beta) (\Delta W_t)^2$$

$$V_0 = 0$$

$$V_1 = 0.9 \times 0 + 0.1 (\Delta W_1)^2$$

$$= 0.1 (\Delta W_1)^2$$

$$V_2 = 0.9 V_1 + 0.1 (\Delta W_2)^2$$

$$= \frac{0.9 \times 0.1 (\Delta W_1)^2}{\quad} + \frac{0.1 (\Delta W_2)^2}{\quad}$$

$$0 < \beta < 1$$

$$\beta = 0.9$$

$$\beta = \frac{1}{1-0.9} = \frac{1}{0.1} = 10$$

$V_t \rightarrow$ never shoots

Disadvantage:-

③ Nothing

Adam (Pse era) $\rightarrow$ RMSProp

Adaptive Moment Estimation

✓ momentum

✓ ad + grad / RMSProp

$$W_{t+1} = W_t - \frac{\eta}{\sqrt{V_t + \epsilon}} \quad \text{momentum}$$

$$\rightarrow V_t = \beta_1 V_{t-1} + (1 - \beta_1) (\Delta W_t)^2 \leftarrow \text{RMSProp}$$

$$\rightarrow m_t = \beta_2 m_{t-1} + (1 - \beta_2) \Delta W_t \leftarrow \text{momentum}$$

$$\boxed{\beta_1 = 0.9 \quad \beta_2 = 0.9}$$

↑ trainable param

$$\boxed{m_0 = 0 \\ V_0 = 0}$$

→ it introduces biases

↓ cancels the effect

$$\hat{m}_t = \frac{m_t}{\epsilon}$$

$$\hat{V}_t = \frac{V_t}{1 - \beta_1^2}$$

$$\hat{m}_t = \frac{m_t}{1 - \beta_2^e}$$

$$V_t = \frac{1}{1 - \beta_1^e}$$

1e #epochs no.
1 2 3

disadvantage : - Nothing